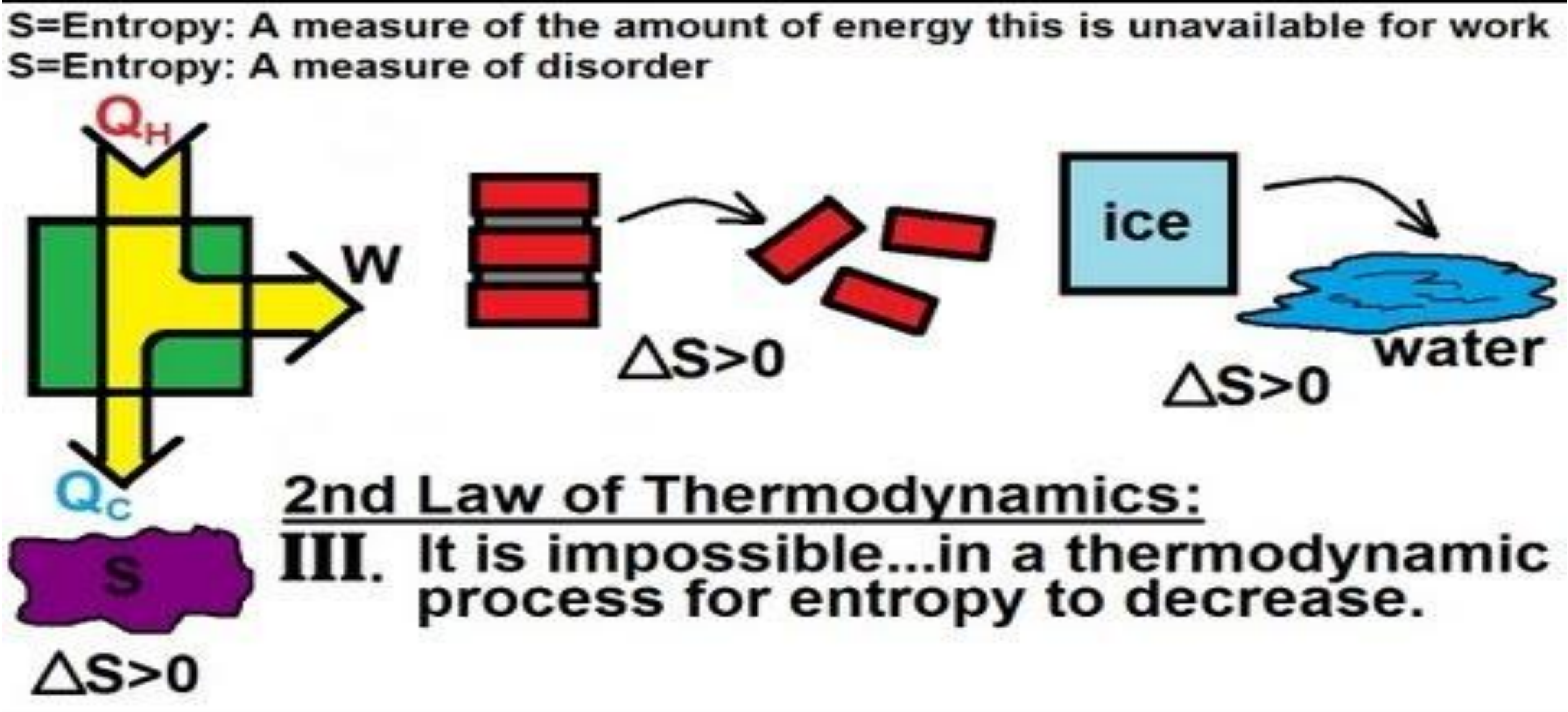

Module-5: Entropy

Thermodynamics (NMEC103) : (Winter 2024-25)

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Every Reversible Process Can be Replaced by a Series of Reversible Adiabatic & Isothermal Process

Applying the 2nd law of thermodynamics it can be shown that , it always possible to replace any reversible process by a series of reversible adiabatic and isothermal process such that the internal energy change, the heat interaction, and the work performed are same, hence a reversible cycle can be replaced by a series of Carnot cycles . Let i-f is any reversible process then i-a is a reversible adiabatic , a-b reversible isothermal and b-f reversible adiabatic such that area under the actual reversible path i-f is same as that under the zigzag path i-a-b-f

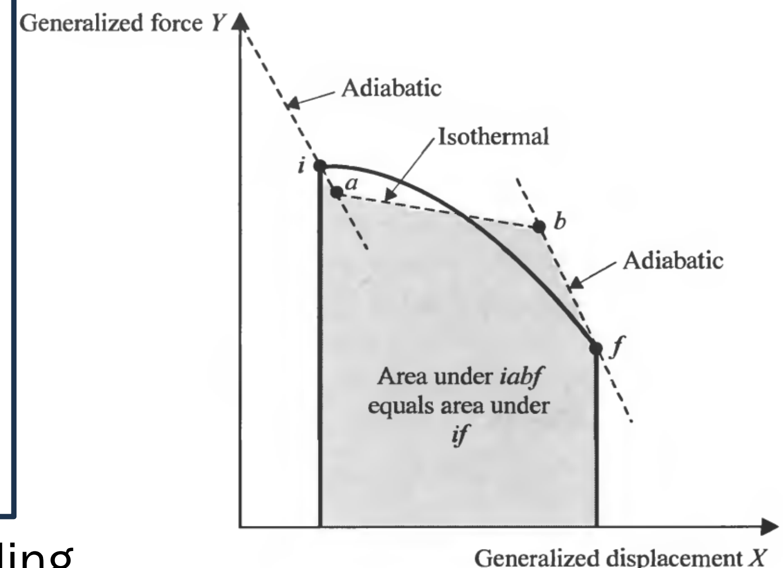
$$Q_{i-f} = U_f - U_i + W_{i-f}$$

$$Q_{i-a-b-f} = U_f - U_i + W_{i-a-b-f}$$

$$Q_{a-b} = U_f - U_i + W_{i-a-b-f}$$

$$\text{if} \quad Q_{i-a-b-f} = Q_{i-f} \Rightarrow W_{i-a-b-f} = W_{i-f}$$

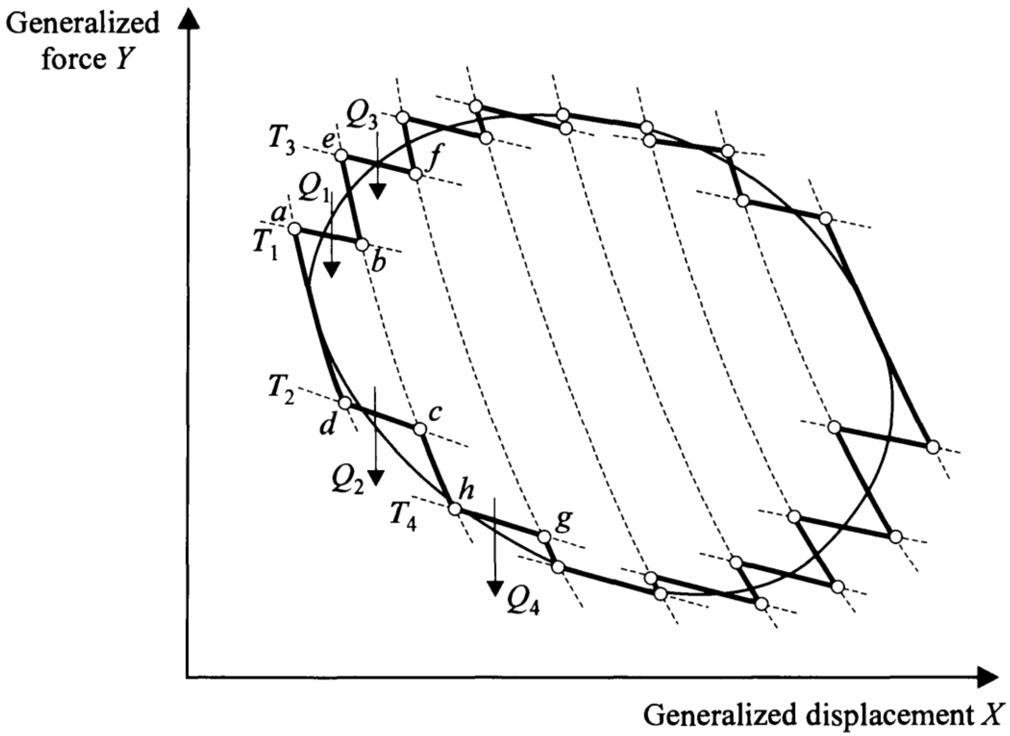
$$\text{if} \quad W_{i-a-b-f} = W_{i-f} \Rightarrow Q_{i-a-b-f} = Q_{i-f}$$



If area under both curve are same then depending on the x and y coordinates the heat/work interactions will be same for both the process

Now for this series of Carnot cycles it can be concluded that

$$\frac{Q_1}{T_1}+\frac{Q_2}{T_2}+\frac{Q_3}{T_3}+\frac{Q_4}{T_4}\dots\dots\dots\dots\dots\dots\dots\dots+\frac{Q_N}{T_N}=0$$



So for any reversible cycle

$$\oint\left(\frac{\delta Q}{T}\right)_{rev}=0$$

$$\int\left(\frac{\delta Q}{T}\right)_{rev} \textit{ is a property}$$

$$\int\left(\frac{\delta Q}{T}\right) \textit{ is not a property}$$

Even for an internally reversible cycle it is true.

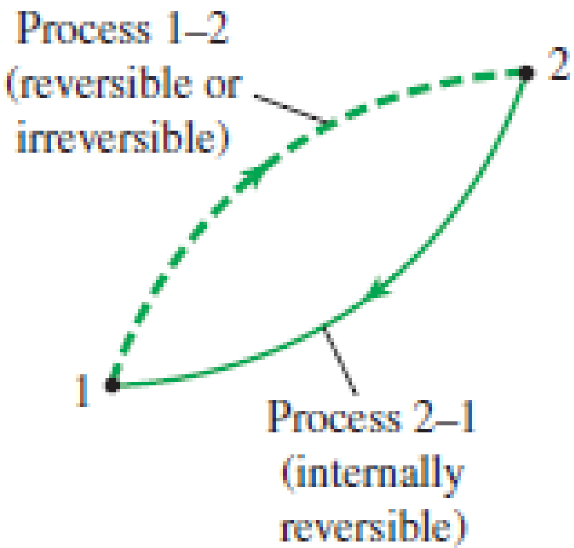
$$\oint\left(\frac{\delta Q}{T}\right)_{\mathrm{int},rev}=0$$

Here δQ is the heat transfer of the system and T is absolute temperature of the system

It is well known that the cyclic integral of a property is zero as it is a point function. So the property entropy is defined as follows

$$\Delta S = \int \left(\frac{\delta Q}{T}\right)_{\text{int},rev} \Rightarrow S_x = S_o + \int_0^x \left(\frac{\delta Q}{T}\right)_{\text{int},rev}$$

Consider a cycle that comprises of two process: 1-2 (reversible/irreversible) and 2-1 internally reversible



From Clausius inequality $\oint \frac{\delta Q}{T} \leq 0$

$$\int_1^2 \frac{\delta Q}{T} + \int_2^1 \left(\frac{\delta Q}{T}\right)_{\text{int},rev} \leq 0$$

$$\int_1^2 \frac{\delta Q}{T} + S_1 - S_2 \leq 0 \Rightarrow S_2 - S_1 \geq \int_1^2 \frac{\delta Q}{T}$$

$$dS \geq \frac{\delta Q}{T}$$

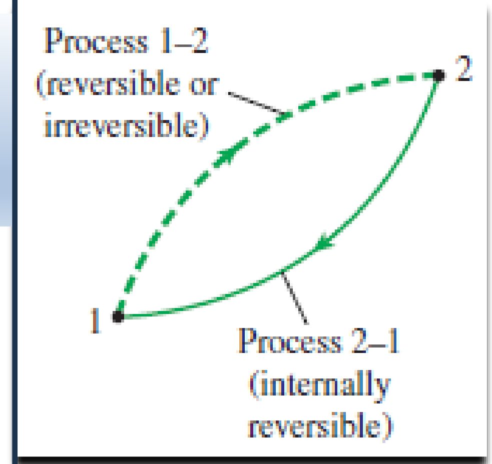
In the differential Form: $dS \geq \frac{\delta Q}{T}$ or $\delta Q \leq TdS$

The above inequality can be looked at in two different way.

For the same end states:

$$\Delta S_{\text{int},rev} = \Delta S_{irrev} = \int_1^2 \left(\frac{\delta Q}{T}\right)_{int,rev}$$

$$\int_1^2 \left(\frac{\delta Q}{T}\right)_{\text{int},rev} > \int_1^2 \left(\frac{\delta Q}{T}\right)_{irrve}$$



For the same δQ and T:

$$\Delta S_{\text{int},rev} < \Delta S_{irrev}$$

ENTROPY GENERATION

From the previous consideration it can be concluded that the **entropy change in a irreversible process is more then that of a reversible process for the same δQ and T**. So the above inequality can be written in an equality form as

$$dS = \frac{\delta Q}{T} + \delta S_{gen} \ldots\ldots\ldots (1)$$

The above equation clearly demonstrates that unlike energy, entropy is not a conserved quantity, energy is conserved but entropy is generated. The entropy generation term which is a path function, is **always positive**, and is a **quantitative measure of the irreversibility** associated with any process, the irreversibility may be due to internal friction, unrestrained expansion, any type of chemical reaction, mixing of fluids, inelastic deformation, etc. or it may be due heat transfer through finite temperature difference within the system or between the system and surrounding. As the **entropy generation** depends on the irreversibility present in the process it is a path function, hence **an inexact differential.**

$$\delta S_{gen} \geq 0 \ldots\ldots\ldots (2)$$

For a reversible process for which entropy generation is zero

$$\delta Q = T ds \,\ldots..(3)$$

Further for a simple compressible closed system

$$\delta W = Pdv \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(4)$$

Putting 3 and 4 for in 1st law for a simple compressible closed system

$$\delta Q = dU + \delta W \,\ldots.(a)$$

$$TdS = dU + pdV = dH - pdV - Vdp + pdV = dH - Vdp \ldots(b)$$

$$TdS = dU + pdV \qquad\qquad TdS = dH - Vdp$$

For an irreversible process with nonzero entropy generation

$$dS = \frac{\delta Q_{irrv}}{T} + \delta S_{gen} \ldots\ldots..(1) \quad \delta Q_{irrv} = TdS - T\delta S_{gen} \ldots\ldots\ldots..(5)$$

From first law applied to the irreversible process:

$$\delta Q_{irrv} = dU + \delta W_{irrv} \ldots\ldots\ldots\ldots\ldots\ldots..(6)$$

From thermodynamic property relation which is also valid for an irreversible process $$TdS = dU + PdV \;......................(7)$$

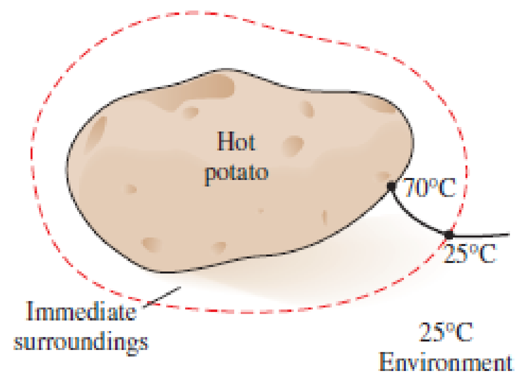
Substituting 5 and 7 in 6

$$\delta Q_{irrv} = dU + \delta W_{irrv} \Rightarrow TdS - T\delta S_{gen} = dU + \delta W_{irrv}$$

$$\Rightarrow dU + PdV - T\delta S_{gen} = dU + \delta W_{irrv} \Rightarrow \delta W_{irrv} = PdV - T\delta S_{gen}$$

The work in an irreversible process is reduced by an amount proportional to the entropy generation. For this reason the term $T\delta S_{gen}$ is known as lost work: $$\delta W_{Lost} = T\delta S_{gen}$$

Any irreversibility during a process occur within the system and its immediate surroundings, and the environment is free of any irreversibilities (generally the environment is assumed to under go a reversible isothermal process). When analysing the cooling of a hot baked potato in a room at 25ºC, for example, the warm air that surrounds the potato is the immediate surroundings, and the remaining part of the room air at 25ºC is the environment. Note that the temperature of the immediate surroundings changes from the temperature of the potato at the boundary to the environment temperature of 25ºC



$$\dot{W}_{Lost} = T_0 S_{gen}$$

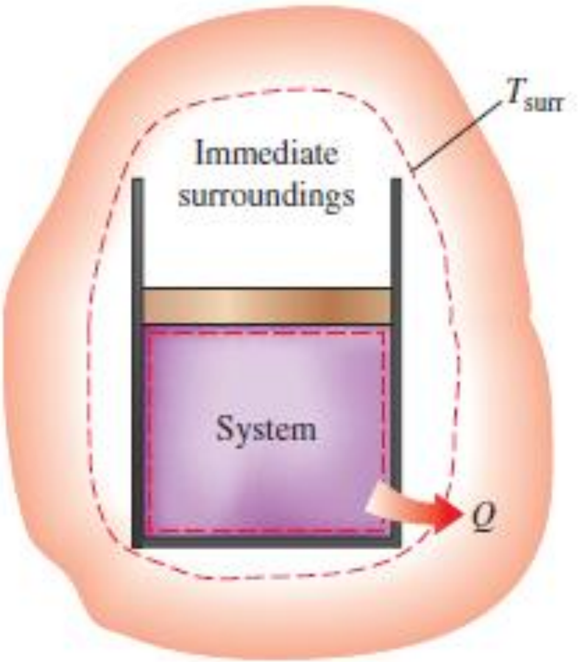
Irreversibility associated with the process

Entropy Balance Equation :

Entropy Change = Entropy Transfer+ Entropy Generation

- Entropy Transfer due to mass Transfer
- Entropy Transfer due to heat transfer

- Because of external and internal irreversibility



Control mass:

$$\frac{dS_{Cm}}{dt} = \sum \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen}$$

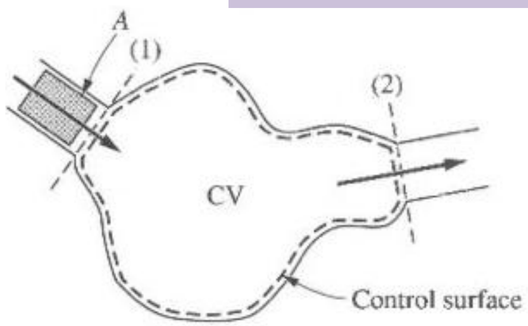
$$\frac{d}{dt}\int_V \rho s dV = \int_A \left(\frac{\dot{q}}{T}\right)_b dA + \dot{S}_{gen}$$

$$S_2 - S_1 = \int_\tau \int_A \left(\frac{\dot{q}}{T}\right)_b dA\, dt + \left(S_{gen}\right)_{1-2}$$

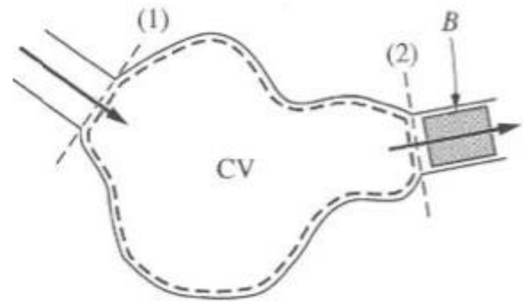
Entropy Balance Equation for a control volume

The entropy balance for any closed system has been shown to be

$$\frac{dS_{Cm}}{dt} = \sum \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen} \ldots\ldots(1)$$



At time t



At time t+Δt

The extension of this equation to a control volume may be made as follows. Let at time t the control mass lies in the within the small region A plus the control region CV. At time $t + dt$ the mass of interest lies in region CV plus a small region B. At these two times the entropies of the control mass are given by

$$S_{Cm,t} = S_A + S_{CV,t} \ldots\ldots\ldots\ldots\ldots\ldots(2)$$

$$S_{Cm,t+\Delta t} = S_B + S_{CV,t+\Delta t} \ldots\ldots\ldots\ldots\ldots (3)$$

Subtracting (2) from (3), yields the change in entropy of the system during that time Δt

$$S_{Cm,t+\Delta t} - S_{Cm,t} = S_{CV,t+\Delta t} - S_{CV,t} + S_B - S_A \ldots\ldots\ldots (4)$$

$$S_{Cm,t+\Delta t} - S_{Cm,t} = S_{CV,t+\Delta t} - S_{CV,t} + S_B - S_A \text{.........................}(4)$$

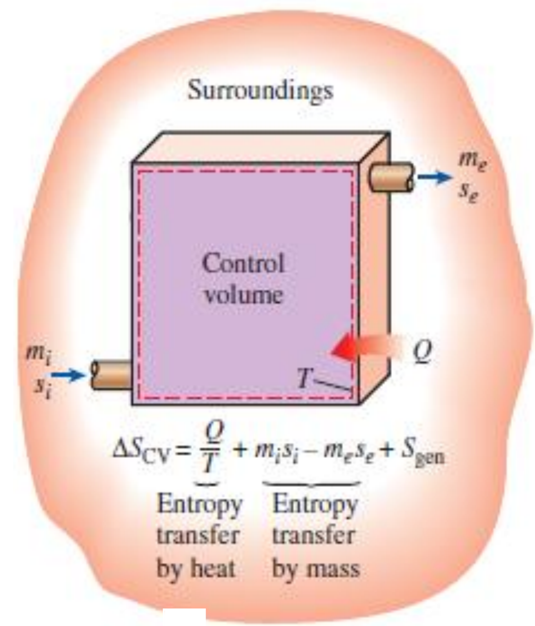
To express this equation on time rate basis we first divide by Δt and replace S_B and S_A as:

$$S_A = m_1 s_1 \qquad S_B = m_2 s_2$$

$$\frac{S_{Cm,t+\Delta t} - S_{Cm,t}}{\Delta t} = \frac{S_{Cv,t+\Delta t} - S_{Cv,t}}{\Delta t} + \frac{m_2 s_2}{\Delta t} - \frac{m_1 s_1}{\Delta t} \text{........}(5)$$

In the limit as Δt approaches zero, the boundary of the control mass and control volume coincide and the above equation becomes

$$\frac{ds_{cm}}{dt} = \frac{ds_{cv}}{dt} + \dot{m}_2 s_2 - \dot{m}_1 s_1 \text{...............} (6)$$

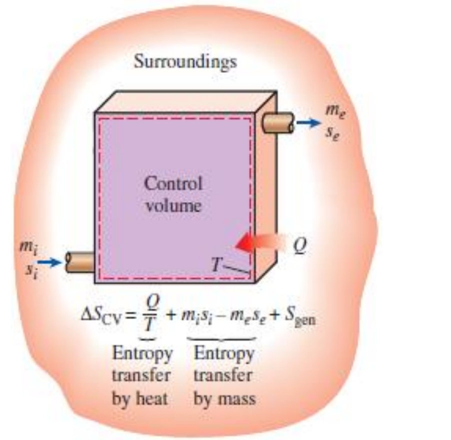


For control volume with multiple inlet and outlets

$$\frac{ds_{cm}}{dt} = \frac{ds_{cv}}{dt} + \sum_{out} \dot{m}_2 s_2 - \sum_{in} \dot{m}_1 s_1 \dots\dots\dots (7)$$

Substituting equation-1

$$\frac{dS_{Cm}}{dt} = \sum \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen} \ldots\ldots(1)$$



$$\frac{ds_{cm}}{dt} = \frac{ds_{cv}}{dt} + \sum_{out} \dot{m}_2 s_2 - \sum_{in} \dot{m}_1 s_1 = \sum \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen} \ldots\ldots(8)$$

$$\frac{ds_{cv}}{dt} = \sum \frac{\dot{Q}_k}{T_k} + \dot{S}_{gen} + \sum_{in} \dot{m}_2 s_2 - \sum_{out} \dot{m}_1 s_1 \ldots\ldots\ldots\ldots(9)$$

$$\frac{d}{dt}\int_V \rho s dV = \sum_i \left(\int_A s\rho \mathrm{V_n} dA\right)_i - \sum_e \left(\int_A s\rho \mathrm{V_n} dA\right)_e + \int_A \left(\frac{\dot{q}}{T}\right)_b dA + \dot{S}_{gen}$$

Entropy production in cyclic devices: $\Delta S = 0$ $\quad \dot{S}_{gen} = -\sum \frac{\dot{Q}_k}{T_k}$

No mass in and out

Useful relationship among properties: The T-ds relation

By combining the definition of entropy with the first law, we now derive a highly useful relationship that is often used for the calculation of entropy and for other purposes.

From the definition of entropy $\Delta S = \int \left(\frac{\delta Q}{T}\right)_{\text{int},rev}$

From 1st law applied to simple compressible closed system undergoing reversible process

$$\delta q = du + \delta w = \delta q_{rev} = du + pdv$$

It can be shown that the above equation is also valid for a reversible process undergone by a steady flow system

From steady flow energy equation: $\delta q = \delta w + dh + dke + dpe = \delta w + du + d(pv) + dke + dpe$

For a reversible process:

$$\delta q_{rev} = -vdp - dke - dpe + du + pdv + vdp + dke + dpe = du + pdv$$

Notice for steady flow process: $pdv \neq \delta w$

$$\Delta S = \int \left(\frac{\delta Q}{T}\right) = \int \frac{du + pdv}{T}$$

$$ds = \frac{du + pdv}{T} \Rightarrow Tds = du + pdv = du + d(pv) - vdp = dh - vdp$$

$$\Delta S = \int \left(\frac{\delta Q}{T}\right)_{\text{int},rev} \quad .(1) \quad Tds = du + pdv \dots (2) \quad Tds = dh - vdp ..(3)$$

The above two equations (2&3) involves only properties, since the functional relation between T, p, u, v, and s is independent of any process. ***So the above equation is no way restricted to reversible process.*** *This, of course also means that the relation holds for a fluid flowing through an open system as well as for a closed system* because entropy is a function of properties such as p, v, t and u and is independent of the position or the state of motion. The integration of right hand side of the equation 1 must be performed for some reversible process between the end states, However, equation 2 involves only properties, and since the functional relationship among p, v, t, u and s is independent of the process. This equation does require a continuous functional relationship among the properties involved and this is turn require that we deal with only equilibrium states. ***This equation is restricted to simple compressible pure substance***.(***If mixing or chemical reaction occur entropy can change even internal energy and volume are held constant; so obviously this equation does not apply***) ***If the effect of electricity, magnetic effect, solid distortion effect, surface tension effect are included (No more a simple system)***

$$\boxed{Tds = du + pdv - \gamma dA - \tau dL - \Im dZ}$$

Equations & Restrictions

(a) $\delta q = du + \delta w$ This is a statement of , first law , appllicable to any stationary closed system

(b) $\delta q = du + pdv$ This is statement of 1^{st} law , restricted to a reversible process of a simple *compressible stationary* closed system.
(This equation can also be applied to a reversible process for a fluid flowing steadily through an open system, but then $pdv \neq \delta w$).
However
the system must be simple compressible

(c) $Tds = du + \delta w$ This is statement combining 1^{st} and 2^{nd} laws and is restricted to a reversible process of a closed stationary system.($Tds = \delta q$)

(d) $Tds = du + pdv$ This is a relationship among properties (p,v,T,s,u) valid for all process between equilibrium states. It is based on 1^{st} and 2^{nd} laws, but in itself it is a statement of neiher. In general, Tds $\neq$ δq and and pdv $\neq$ δw by comparison with equation a it is seen, in fact that for a closed system, Tds = δq only when pdv = δw and vice versa

But the system should be a simple compressible pure substance

Illustrating the difference between equation “a” and “d”

$$\delta q = du + \delta w \quad \ldots .(a) \qquad Tds = du + pdv \quad \ldots ..(d)$$

Consider the system composed of gas trapped in a closed rigid and thermally insulated vessel fitted with a paddle wheel , so that work can be done on the gas. Let the paddle wheel is turned by external motor. Notice that

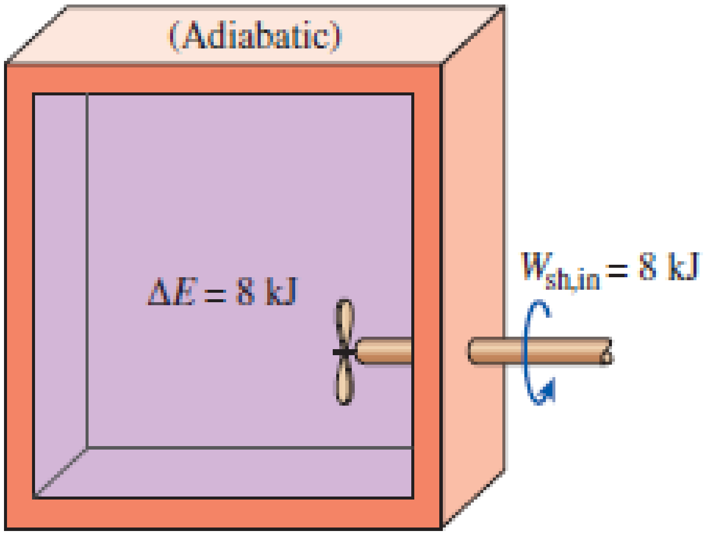
pdv =0 rigid vessel –No change in Volume

but $\delta W \neq 0$

$\delta Q = 0$ (insulated vessel) but $Tds \neq 0$

Equation “a” reduces to $0 = du + \delta w$

Equation “d” reduces to $Tds = du + 0$



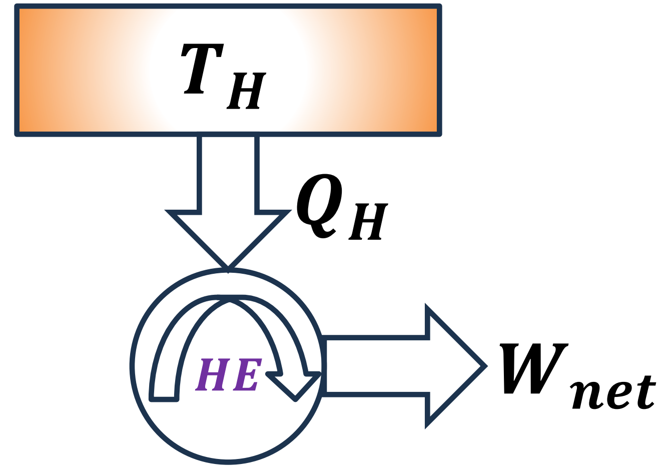
The above demonstration clearly shows that equation a and d is significantly different

Check for Violation of 2nd law from Entropy Principle

Case:1

$$(\Delta S)_{reservoir} = \frac{-Q_H}{T_H}$$

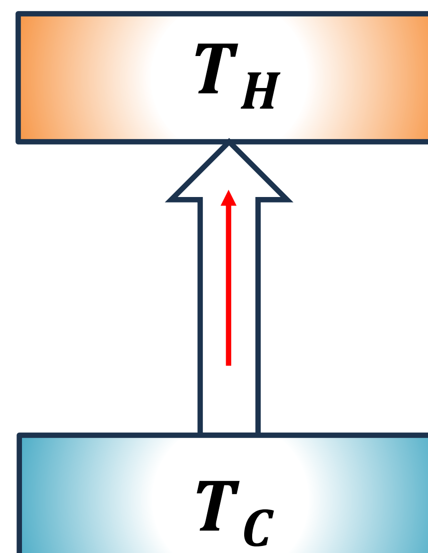
$$(\Delta S)_{Heat\ engine} = 0$$



$$S_{gen} = \Delta S_{total} = (\Delta S)_{reservoir} + (\Delta S)_{Heat\ engine}$$
$$= \frac{-Q_H}{T_H} + 0 < 0$$

Case:2

"Heat can not be caused to flow from a cooler to hotter body without producing some other effect" If the excluded process is possible, we would have heat extracted from a body at T_C, causing an entropy change $-Q/T_C$ and added to a body at T_H, causing an entropy change $-Q/T_H$ then we would have $S_{gen} = \Delta S_{total} = \frac{-Q}{T_C} + \frac{Q}{T_H} = Q\left(\frac{1}{T_H} - \frac{1}{T_C}\right) = Q\left(\frac{T_C - T_H}{T_H T_C}\right) < 0 \because T_C < T_H$



Application of 2nd Law of Thermodynamics to Steady Flow Devices

Nozzles, Diffusers, and Throttles: are normally steady state steady flow, single inlet, single-outlet open system with approximately constant surface temperature, and that they may or may not be adiabatic

For adiabatic nozzles, diffusers and throttle valve

$$\dot{S}_{gen} = \dot{m}(s_{out} - s_{in})$$

For isothermal surface nozzles, diffusers and throttle valve

$$\dot{S}_{gen} = \dot{m}(s_{out} - s_{in}) - \frac{\dot{Q}}{T}$$

$$Tds = dh - vdp$$

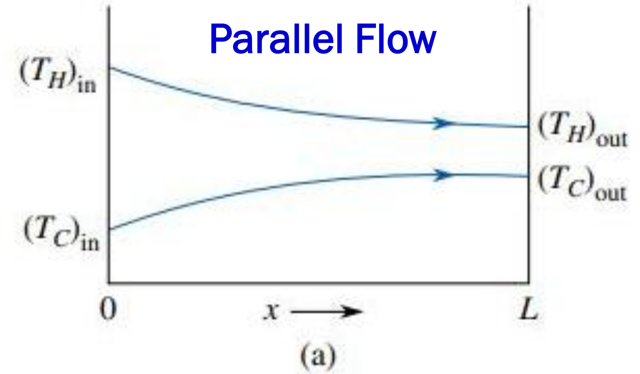
For throttling process

$$Tds = -vdp \because dh = 0$$

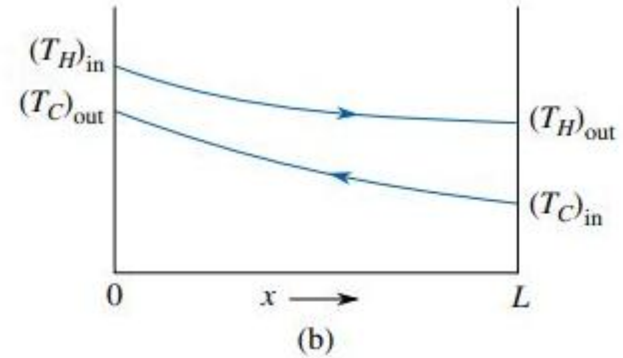
$$ds = S_{gen} = \frac{-vdp}{T} > 0 \because dp < 0 \text{ for throttling}$$

Heat Exchangers: The heat exchangers are normally classified as either parallel flow, counterflow or cross flow

Parallel Flow



(a)



(b)

Counter Flow

$$\dot{Q} = UA(\Delta T)_{LMTD}$$

$$(\Delta T)_{LMTD} = \frac{(T_H - T_L)|_{x=L} - (T_H - T_L)|_{x=0}}{ln[(T_H - T_L)|_{x=L}/(T_H - T_L)|_{x=0}]}$$

$$\dot{m}_H(h_{in} - h_{out})_H = \dot{m}(h_{out} - h_{in})_C$$

$$\dot{S}_{gen} = \dot{m}_H(s_{out} - s_{in})_H + \dot{m}_C(s_{out} - s_{in})_C$$

Incompressible Fluid

$$\dot{S}_{gen} = \dot{m}_H c_H ln\left(\frac{T_{out}}{T_{in}}\right)_H + \dot{m}_C c_C ln\left(\frac{T_{out}}{T_{in}}\right)_C$$

Ideal Gas

$$\dot{S}_{gen} = \dot{m}_H\left[\left(c_p\right)_H ln\left(\frac{T_{out}}{T_{in}}\right)_H - R_H ln\left(\frac{p_{out}}{p_{in}}\right)_H\right] + \dot{m}_C\left[\left(c_p\right)_C ln\left(\frac{T_{out}}{T_{in}}\right)_C - R_C ln\left(\frac{p_{out}}{p_{in}}\right)_C\right]$$

Shaft Work Machines & Entropy Production

Are any device that has work or power input or output through a rotating or reciprocating shaft. These devices are generally steady state, steady flow, single inlet and outlet systems.

$$\dot{Q} - \dot{W} = \dot{m}\left((h_2 - h_1) + \left(\frac{V_2^2}{2} - \frac{V_1^2}{2} \right) + g(Z_2 - Z_1) \right)$$

Assuming isothermal system boundary

$$\dot{m}(s_1 - s_2) + \frac{\dot{Q}}{T_b} + \dot{S}_{gen} = 0$$

$$\dot{m}T_b(s_1 - s_2) + T_b\dot{S}_{gen} = -\dot{Q}$$

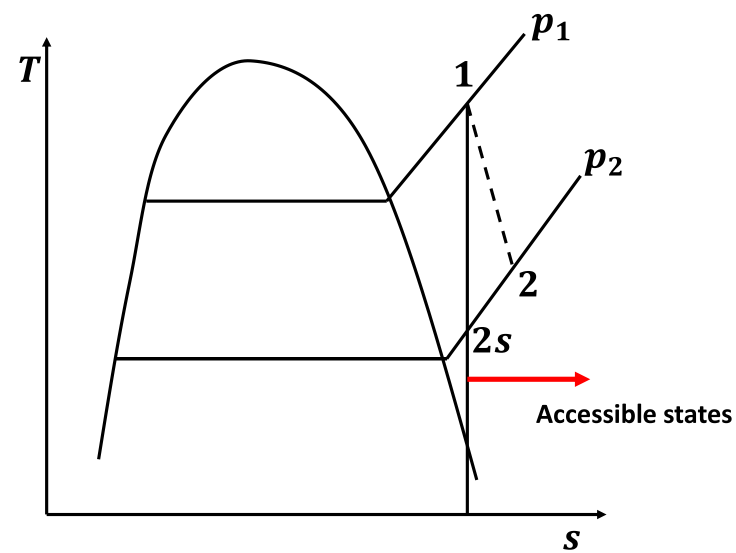
$$-\left(\dot{m}T_b(s_1 - s_2) + T_b\dot{S}_{gen}\right) - \dot{m}\left((h_2 - h_1) + \left(\frac{V_2^2}{2} - \frac{V_1^2}{2} \right) + g(Z_2 - Z_1) \right) = \dot{W}$$

Isentropic Efficiency (is a measure of effectiveness of real adiabatic process relative to best possible)of Turbine, Compressor/Pumps and Nozzles/ Diffusers

$1 - 2\ adiabatic; 1 - 2s\ isentropic$

Isentropic efficiency of Turbines:

$$\eta_{is,T} = \frac{w_{T,ad}}{w_{T,is}} = \frac{(h_1 - h_2)_{ad}}{(h_1 - h_{2s})}$$

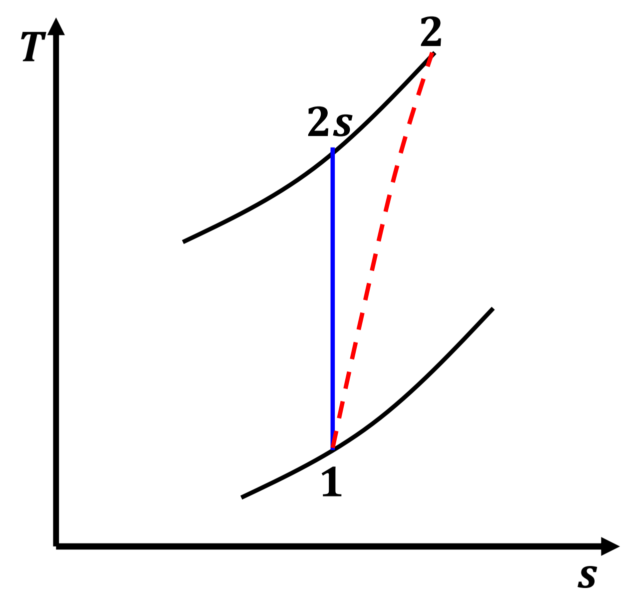


Isentropic efficiency of Nozzle:

$$\eta_{is,Noz} = \frac{\left(V_{2,ad}^2 - V_1^2\right)/2}{(V_{2s}^2 - V_1^2)/2} = \frac{(h_1 - h_2)_{ad}}{(h_1 - h_{2s})}$$

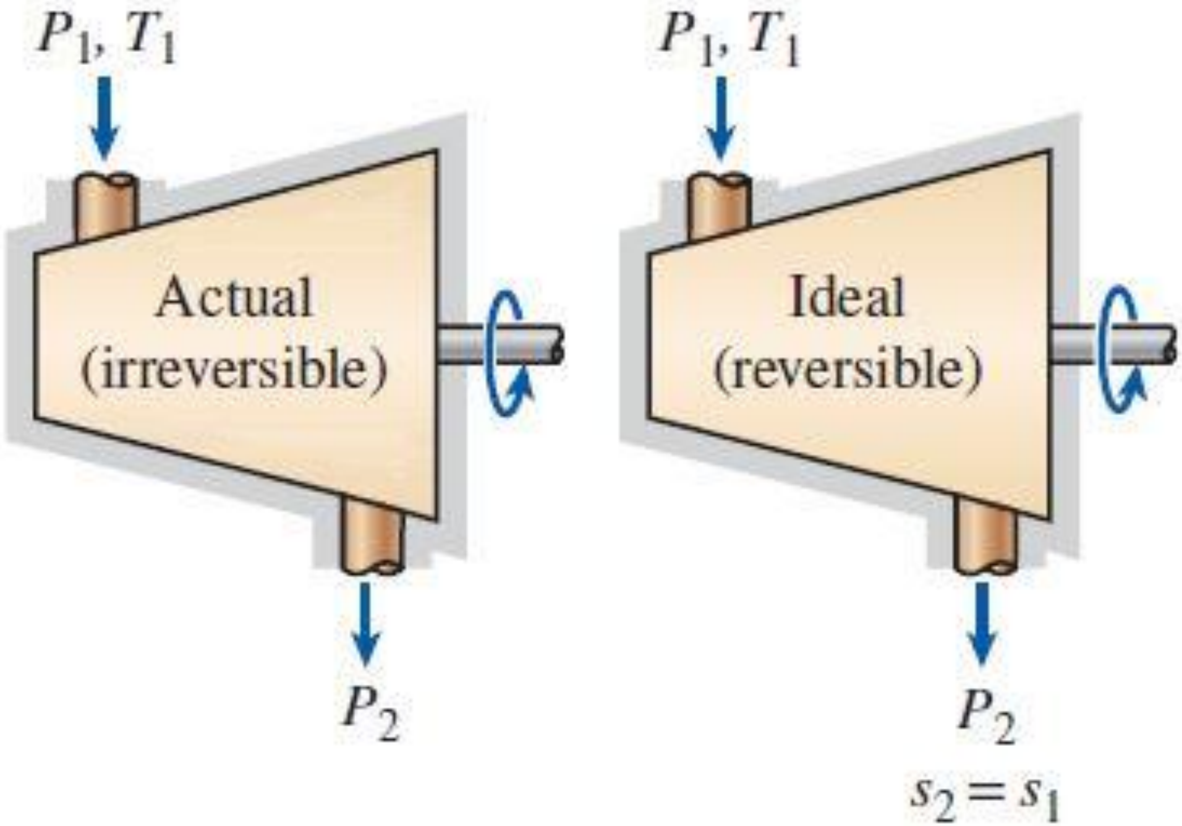
Isentropic efficiency of Compressor/Pump:

$$\eta_{is,C} = \frac{w_{C/P,is}}{w_{C/P,ad}} = \frac{h_{2s} - h_1}{(h_2 - h_1)_{ad}}$$



Isentropic efficiency of Diffuser:

$$\eta_{is,Diff} = \frac{h_{2s} - h_1}{(h_2 - h_1)_{ad}} = \frac{V_{2s}^2 - V_1^2}{V_2^2 - V_1^2}$$


P_1, T_1
Actual
(irreversible)
P_2
P_1, T_1
Ideal
(reversible)
P_2
$s_2 = s_1$